The role of cognateness in native spoken word recognition

Introduction

When some German speakers listen to the song *The Power* by SNAP! (*Coyote Ugly*, 2000) many of them mishear the line "I've got the power" as Agatha Bauer. This auditory illusion, aptly termed as Soramimi (lit. "empty ear") in Japanese, is a particular case of homophonic translation, where words or phrases from one language are perceived as similar-sounding words and phrases in a different language, without necessarily preserving the meaning (Gasparov, 2006). The phenomenon of Soramimi is common across languages and cultures (Dembeck, 2015; Efimova et al., 2018) and gives insight into how unfamiliar phonological forms are processed by language speakers, with implications for theories of spoken word recognition and practical applications for the acquisition of second languages. In the present study, we examined how homophonic translation allows listeners with no prior experience with the presented language to identify the correct translations of foreign words with varying levels of similarity to words in the monolingual listeners' native language.

Homophonic translation provides insight into how listeners map spoken input onto native-language lexical representations even when no exact lexical match is available. By examining homophonic translation, we aim to better understand how native-language lexical competition may interfere with word recognition in a second language, a challenge commonly faced by bilingual speakers. Our study extends on previous work which characterised how the phonological properties of two languages interact during speech processing by bilinguals with differing levels of familiarity and proficiency with a second language. In bilinguals, lexical competition can arise from words in both languages during word recognition, making it a non-trivial task to disentangle the effects of lexical competition in the native and second language during speech processing. With bilinguals, assumptions about which lexical representations can be activated by the input, which can compete with others, and with which other representation they can compete (e.g., within-language, between-language, or both) impacts

the kind of conclusions that can be extracted (see Costa et al., 2006 for discussion). By testing the mechanisms affecting homophonic translation among listeners with no prior experience with the presented language, we aim to minimise potential confounds from second-language lexical knowledge and more directly investigate how the first-language lexicon influences homophonic translation.

Homophonic Translation: Matching Phoneme Sequences with Lexical Items

Homophonic translation has mostly received attention as a perceptual curiosity or as a literary technique, intentionally employed by translators to preserve the meter and sound structure (e.g., rhymes) of an original text in the language of translation (e.g., Levick, 2016; Pilshchikov, 2016). Little attention has been paid to homophonic translation from a language processing perspective. When considering how homophonic translation can contribute to our understanding of spoken word recognition, it is important to examine what occurs during real-time processing by active listeners of speech. Under theories of lexical activation, multiple possible candidate words are simultaneously activated during the process of spoken word recognition (Marslen-Wilson & Welsh, 1978) and speech production (Levelt et al., 1991), with the level of activation determined by how well a candidate matches the auditory input. In the case of homophonic translation, despite the presented word being in a foreign language, the activated candidates will be words or phrases in the listener's native language as the brain attempts to find the closest equivalent. Three main phonological processes could be used to explain how listeners reconstruct foreign input to generate words in their native language: insertion (e.g., English cry /kɹaɪ̯/ to Japanese kurai /kɯɾai/ [dark]), deletion (e.g., English go /goʊ/ to Japanese go /go/ [Go, a board game]), and alternation (e.g., English low /loʊ/ to Japanese rou /ɾo̞ː/ [wax]) (Otake, 2007). These findings suggests that homophonic translations can be explained, to some extent, as the result of the listeners accommodating the strings of

sounds to their native phonology and phonotactics pe(Dupoux et al., 1999; Peperkamp et al., 2008)

Homophonic Translation in Second Language Learning: Cognates and Competition

Word pairs like German pinguin and English penguin present a special case of homophonic translation where the meaning of the original word or phrase is preserved, allowing the listener to match a foreign word to the correct translation in their native language even if they are completely unfamiliar with the language being spoken. Such translation equivalents with high phonological similarity-known in the bilingual literature as cognates—are common across many languages. Historically-related language like Spanish and Catalan share many form-similar translation equivalents (Schepens et al., 2012), for example puerta and porta (door in Spanish and Catalan, respectively). Second language learners have been found to learn cognates more easily than non-cognates (Hall, 2002; Otwinowska & Szewczyk, 2019). The phonological processes underlying homophonic translation may allow new learners of a second language to recognise the similarities between the second language and their native language, and to capitalise on that phonological similarity to facilitate the learning process.

However, these mechanisms of automatically comparing foreign input to one's native language may also cause interference. The presence of cross-linguistic pairs which are phonologically similar but differ in meaning (e.g., false friends) may act as distractors during lexical access, obstructing the selection of the appropriate target translation in the listener's lexicon (Otwinowska & Szewczyk, 2019). Homophonic translation typically occurs despite the phonological form not mapping perfectly onto lexical items in the listeners' native language. This may make homophonic translation more sensitive to competition from other potential candidates. The phonological neighbourhood density of a given word-form refers to the number of words-forms that differ in a single phoneme from it. For instance, the English word cat belongs to a dense phonological neighbourhood, with 50 English phonological neighbours (e.g., hat, bat, cap), while the word charger belongs to a sparse phonological neighbourhood,

with six neighbours (e.g., charged, larger, charter, charmer) (Marian et al., 2012). Words from dense neighbourhoods are processed, and subsequently recognised, more slowly and less accurately in the auditory modality than words from sparse neighbourhoods (e.g., Dufour & Frauenfelder, 2010; Dufour & Peereman, 2003; Goldinger et al., 1989; Hamburger & Slowiaczek, 1996; Luce et al., 1990; Luce & Pisoni, 1998; Vitevitch & Luce, 1998). Interference effect of phonological neighbourhood density has also been reported not only within a language but also across languages in bilinguals (Weber & Cutler, 2004). Thus, in the case of homophonic translation, interference may come from lexical competition from phonological neighbours in the learner's native language, especially for cases where neighbours share higher phonological similarity with the presented word than the target translation, or are lexically more frequent (Luce & Pisoni, 1998).

The Present Study

We examined the mechanisms of homophonic translation through a translation elicitation task in which participants listened to words from an unfamiliar language (presented words), and then tried to guess the words' translations in their native language (target translations). Since participants were unfamiliar with the presented language, they could only rely on phonological similarity between the presented language and their native language to successfully translate the words, instead of on previously learned word-meaning associations. Listeners of an unfamiliar language provide a unique case to investigate the dynamics of lexical processing involved in homophonic translation. For listeners with no prior experience to the target language, the influence of existing knowledge about its word forms, phonology and meanings can be largely ruled out. As a result, observed effects can be more confidently attributed to the similarity between the features of the presented language and the native language of the participant. and with which other representations We predicted that participants' performance should increase when the translation pairs are phonologically more

similar. A second prediction was that the size of the facilitatory role of phonological similarity would be inversely proportional to the amount of higher-frequency phonological neighbours of the presented word in the target language. Even if the presented word and its target translation share high phonological similarity, participants may still provide incorrect translations if the presented word also shares high phonological similarity with other words in the native language, paralleling the dynamics of lexical selection engaged during native speech processing. To test this second prediction, we calculated a lexical frequency-dependent measure of cross-language phonological neighbourhood density, in which a neighbour is counted only if it is higher frequency and is one phoneme apart from the presented word. If the phonological neighbourhood density of the target translation affects participants' performance negatively, this would suggest that competitors in the native language affect recognition of non-native words during recognition of foreign speech.

Experiment 1

This first study was aimed at exploring the interplay between cross-linguistic similarity and phonological neighbourhood density in homophonic translation of Spanish and Catalan words by British English natives. English, a Germanic language, shares moderate overlap with the Romance languages Spanish and Catalan, and thus we can examine the mechanisms of homophonic translation when the candidate words have some overlap but are not exact matches with the speech input.

Methods

All materials, data, and code used in this study are hosted in the Open Science Framework (https://osf.io/9fjxm/) and a GitHub repository https://github.com/gongcastro/translation-elicitation, along with additional notes. For reproducibility, a Docker image of the RStudio session is available on DockerHub (https://hub.docker.com/r/gongcastro/translation-elicitation).

Participants

We collected data from 71 British English-native adults living in the United Kingdom (*Mean* = 21.76 years, *SD* = 2.15, range = 18-26, 46 female). Data collection took place from June 4th, 2020 to June 25th, 2020. Participants were recruited via Prolific (£5 compensation) and SONA (compensation in academic credits). Participants gave informed consent before providing any data. The study was conducted in accordance with ethical standards of the Declaration of Helsinki and the protocol was by the University of Oxford Medical Sciences Inter-Divisional Research Ethics Committee (IDREC) (R60939/RE007). Participants were asked to complete the experiment using a laptop in a quiet place with good internet connection.

Table 1

Participant information. N *indicates the number of included participants, with excluded participants between parentheses.*

	N	Age: Mean (SD)	Age: Range	Second language
Experiment 1				
spa-ENG	35 (8)	21.8 (2.08)	18–26	Russian (1)
cat-ENG	36 (4)	21.72 (2.25)	18–25	French (1) , German (1) , Italian (1), Punjabi (1), Several (1)
Experiment 2				
cat-SPA	33 (12)	21.85 (3)	18–33	French (9) , German (1) , Italian (2)
Experiment 3				
spa-ENG	32 (1)	21.72 (2.59)	18–26	German (2) , Japanese (1)
cat-ENG	32 (0)	22.31 (2.39)	18–26	Cantonese (1), Irish (1)

Stimuli

We created two lists of words to be presented to participants in the auditory modality: one in Catalan and one in Spanish. Words in the Catalan list were 6.67 phonemes long on average (*SD* = 2.06, range = 2-11), and the orthographic forms of their English translations

(which participants had to type) were 5.12 characters long on average (*SD* = 1.56, range = 3-9). Words in the Spanish list were 7.27 phonemes long on average (*SD* = 2.05, range = 3-13), and the orthographic form of their English translations were 5.29 characters long on average (*SD* = 1.77, range = 3-12).

In each trial, participants listened to one audio file, which contained a single word. Audio files were the same ones used in child experiments conducted in the Laboratori de Recerca en Infància, at Universitat Pompeu Fabra. These audio files were recorded as part of a larger project that this study was part of by a proficient Catalan-Spanish female bilingual from the Metropolitan Area of Barcelona, in a child-directed manner. Catalan and Spanish words were recorded at 44,100 Hz in the same session, and then de-noised using Audacity and normalised at peak intensity using Praat (Broersma & Weenink, 2021). The average duration of the Catalan audio files was 1.24 seconds (*SD* = 0.19, range = 0.8-1.58). The average duration of the Spanish audio files was 1.16 seconds (*SD* = 0.15, range = 0.78-1.53).

We defined three predictors of interest: the lexical frequency of the correct English translation (*Frequency*), the phonological similarity between the presented word and its correct English translation (*Similarity*), and the presented word's number of cross-language phonological neighbours (*CLPN*) in English. *Frequency* was included as a nuisance predictor, under the hypothesis that participants would provide correct responses more often when the correct translations are high frequency words in their native language, keeping other predictors constant. Lexical frequencies of correct English and Spanish translations were extracted from SUBTLEX-UK (Van Heuven et al., 2014) and SUBTLEX-ESP (Cuetos et al., 2011), respectively, and transformed to Zipf scores. Translations without a lexical frequency value were excluded from data analysis (2 in the Catalan list, 5 in the Spanish list). *Similarity* was calculated as the Levenshtein similarity between the X-SAMPA transcriptions of each pair of translations using the `stringdist` R package (van der Loo, 2014). The Levenshtein distance computes the edit distance between two character strings—in this case, two phonological

transcriptions—by counting the number of additions, deletions, and substitutions necessary to make both strings identical (Levenshtein, 1966). We divided this edit distance by the number of characters in the longest X-SAMPA transcription of the translation pair. This transformation accounts for the fact that Levenshtein distance tends to increase with the length of the transcriptions. For interpretability, we subtracted this proportion from one, so that values closer to one correspond to higher similarity between phonological transcriptions, and values closer to zero correspond to lower similarity between phonological transcriptions. For example, the *table* (`teIb@l`)-*mesa* (`mesa`) translation pair had 17% similarity, while the *train* (`trEIn`)-*tren* (`tREn`) translation pair had 60% similarity. We calculated the number of *CLPN* for each Catalan and Spanish word by counting the number of English words with same or higher lexical frequency, and whose phonological transcription (in X-SAMPA format) differed by up to one phoneme from that of the presented Catalan or Spanish word. Lexical frequencies and phonological transcriptions were extracted from the multilingual database CLEARPOND (Marian et al., 2012). These phonological transcriptions were generated from eSPEAK http://espeak.sourceforge.net/. Table 2 summarises the lexical frequency, *CLPN* and phonological overlap of the words included in the Catalan and the Spanish lists.

Table 2

Stimuli information. M*: mean,* SD*: standard deviation,* Min*: minimum,* Max*: maximum.*

	Lexical frequency				Phonological similarity				CLPN			
	M	SD	Min	Max	M	SD	Min	Max	M	SD	Min	Max
spa-ENG	5.88	0.57	4.43	7.24	0.13	0.18	0	0.75	0.20	0.81	0	5
cat-ENG	5.89	0.59	4.43	7.27	0.17	0.19	0	0.67	0.78	2.55	0	15
cat-SPA	5.79	0.63	4.48	7.70	0.39	0.26	0	1.00	0.88	2.09	0	12

Procedure

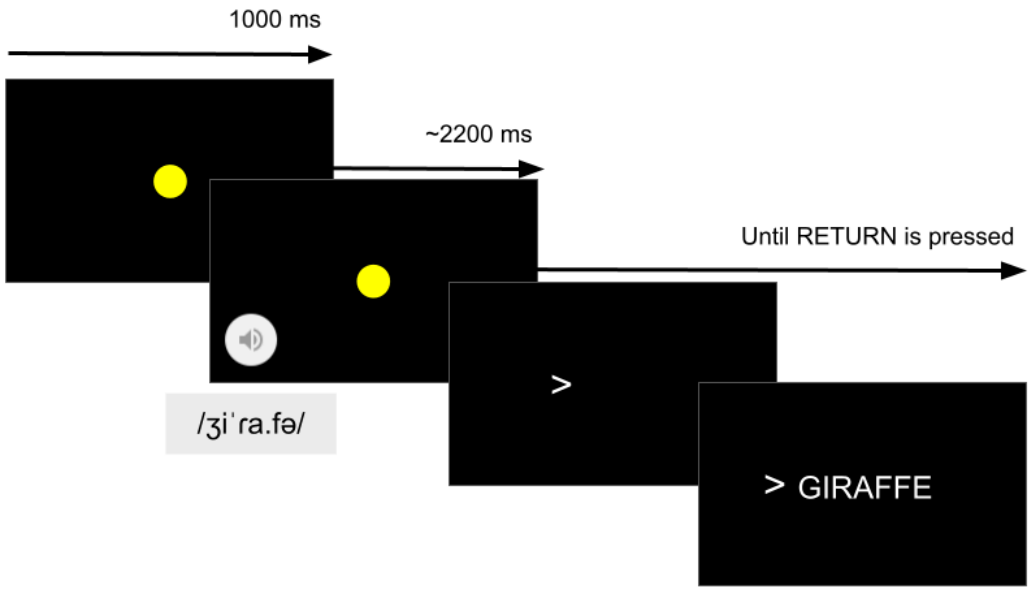
The experiment was implemented online using Psychopy/Pavlovia (Peirce et al., 2019). Participants accessed the study from a link provided by Prolific or SONA and completed the

experiment from an internet browser (Chrome or Mozilla). After giving their consent for participating, participants answered a series of questions about their demographic status, their language background, and the set up they were using for completing the study. Then, participants completed the experimental task. Participants were informed that they would listen to a series of pre-recorded words in either Catalan or Spanish. They were instructed to listen to each word, guess its meaning in English, and type their answer as soon as possible. Participants were randomly assigned to the Catalan or Spanish lists. Participants in the Catalan list were presented with 83 trials, and participants in the Spanish list were presented with 99 trials.

Figure 1 illustrates the time course of a trial in the translation elicitation task.Each trial started with a yellow fixation point presented during one second on the centre of the screen over a black background. After one second, the audio started playing while the dot remained being displayed until the audio ended. Upon the offset of the fixation point and audio, participants were prompted to write their answer by a “>” symbol. Typed letters were displayed on the screen in real time to provide visual feed-back to participants. Participants were allowed to correct their answer. Then, participants pressed the `RETURN/ENTER` key to confirm their answer and start a new trial. The task took approximately 15 minutes to complete.

Figure 1

Schematic representation of a trial in the experimental task. The trial stars with the presentation of a fixation point in the centre of the screen (yellow dot). After 1,000 ms, the auditory word was presented while the fixation point remained on screen. After the offset of the audio, the fixation point disappeared and a visual prompt (>) was presented. Participants then wrote their answer and clicked `RETURN`*, making the end of the trial.*



Data analysis

Data processing. After data collection, participants' answers were manually coded into the following categories: *Correct*, *Typo*, *Wrong*, *Other*. A response was coded as *Correct* if the provided string of characters was identical to the orthographic form of the correct translation. A response was coded as *Typo* if the participant provided a string of characters that was only one edit distance (addition, deletion, or substitution) apart from the orthographic form of the correct translation (e.g., "pengiun" instead of "penguin"), as long as the response did not correspond to an extant English word. Responses not meeting the criteria for previous categories were labelled as *Wrong* or *Other* (see Data analysis section for more details). Both

Correct and *Typo* responses were considered as correct, while *Wrong* responses were considered as incorrect. *Other* responses were excluded from data analysis. Trials in which participants took longer than 10 seconds to respond were also excluded.

Modelling approach and statistical inference. We modelled the probability of participants guessing the correct translation of each presented word using a generalised multilevel Bayesian regression model with a Bernoulli logit link distribution. We included as fixed effects the intercept, the main effects of *Frequency*, *Similarity*, *CLPN*, and *Group* (sum-coded as `cat-ENG = -0`, `spa-ENG = +0.5`, Schad et al., 2020) and the three-way interaction between *Similarity*, *CLPN*, and *Group*. The predictor *Group* was included to account for any differences in task performance between English participants translating Catalan and Spanish words. We also included participant-level random intercepts and slopes for the main effects and the interaction. Equation 1 shows a formal description of the model. Numeric predictors were standardised before entering the model by subtracting the mean and dividing by the standard deviation.

$$
\begin{aligned}
&\textbf{Likelihood} \\
y_i &\sim \text{Bernoulli } (p_i) \\
&\textbf{Parameters} \\
\text{Logit } (p_i) &= \beta_{0[p,w]} + \beta_{1[p]}\text{Frequency}_i + \beta_{2[p]}\text{Similarity}_i + \beta_{3[p]}\text{CLPN}_i + \beta_{4[p]}\text{Group}_i + \\
&\quad \beta_{5[p]}(\text{Similarity}_i \times \text{CLPN}_i) + \beta_{6[p]}(\text{Similarity}_i \times \text{Group}_i) + \\
&\quad \beta_{7[p]}(\text{CLPN}_i \times \text{Group}_i) + \beta_{8[p]}(\text{Similarity}_i \times \text{CLPN}_i \times \text{Group}_i) \\
\beta_{0-8[p,w]} &\sim \mathcal{N}(\mu_{\beta_j}, \sigma_{\beta_j}) \text{ , for participant } p \text{ in } 1, ..., P \text{ and word } w \text{ in } 1, ..., W \\
\beta_{1-8[p]} &\sim \mathcal{N}(\mu_{\beta_j}, \sigma_{\beta_j}) \text{ , for participant } p \text{ in } 1, ..., P \\
&\textbf{Prior} \\
\mu_{\beta_{p,w}} &\sim \mathcal{N}(0, 0.1) \\
\sigma_{\beta_{p,w}} &\sim \text{Exponential } (3) \\
\rho_{p,w} &\sim \text{LKJCorr } (5)
\end{aligned}
\tag{1}
$$

We followed Kruschke and Liddell (2018)'s guidelines for statistical inference. We first specified a region of practical equivalence (ROPE) around zero ([−0.1, +0.1], in the logit scale). This area indicates the values of the regression coefficients that we considered equivalent to zero. We then computed the 95% posterior credible intervals (CrI) of the regression coefficients of interest, which indicates the most likely range of values that contains the true value of the coefficient with 95% probability. Finally, for each regression coefficient we calculated the proportion of the 95% CrI that fell inside the ROPE, noted as *p*(ROPE). This proportion indicates the probability that the true value of the coefficient is equivalent to zero. For instance, if the *p*(ROPE) of a regression coefficient β is 0.5, this indicates that there is a 50% probability that the true value of β is zero or equivalent to zero, given the data. If *p*(ROPE)=0.01, this indicates that there is a 1% probability that the true value of the regression coefficient is zero or

equivalent. If p(ROPE)=0.99, this indicates that there is a 99% probability that the true value of the regression coefficient is zero or equivalent.

All analyses were performed in the R environment (R Core Team, 2013). We used the tidyverse family of R packages (Wickham et al., 2019) to process data and to generate figures. We used the `brms` R package (Bürkner, 2017) using the `cmdstanr` back-end to the Stan probabilistic language (Carpenter et al., 2017) to estimate and compare the models. Model diagnostics and further detytails can be found in the Supplementary Materials.

Results and discussion

We collected data for a total of 6,446 trials completed by 72 participants. We excluded trials in which participants did not enter any text (n = 72), in which a response in a language other than English was provided (e.g., "`agua`", n = 51), in which participants did not provide a whole word (e.g., "`f`", n = 5), and in which participants added comments to the experimenter (e.g., "`unsure`", n = 13). In addition, we excluded data from participants that self-rated their oral and/or written skills in Catalan and Spanish, or any other second language as four or higher in a five-point scale (n = 2), were diagnosed with a language disorder (n = 1), or did not contribute more than 80% of valid trials (n = 9).

After applying trial-level and participant-level exclusion criteria, the resulting dataset included 5,204 trials provided by 54 participants. Of those trials, 2,602 were provided by 27 participants who listened to Catalan words, and 2,604 trials were provided by 32 participants who listened to Spanish words. Responses given by English participants to Catalan presented words were 5.35 characters long on average (*Median* = 5, *SD* = 1.79, range = 1-14), while their translations to Spanish responses were 5.57 characters long on average (*Median* = 5, *SD* = 1.97, range = 2-21).

Table 3 shows a summary of participants' accuracy across Experiments 1, 2, and 3. Participants translating Catalan words and participants translating Spanish words performed equivalently, as indicated by the regression coefficient of *Group* (β = −0.171, 95% CrI = [−0.474,

0.145], p(ROPE) = 0.277). Overall, participants responded less accurately to words with more CLPNs than to words with fewer CLPNs, regardless of the amount of phonological similarity between the presented word and its translation. This is indicated by the size of the regression coefficient of the two-way interaction between *Similarity* and *CLPN* (β = −0.233, 95% CrI = [−0.366, −0.112], p(ROPE) = 0.017). As anticipated, participants' performance benefited from an increase in *Similarity* (β = 1.437, 95% CrI = [1.323, 1.552], p(ROPE) = 0), while the number of *CLPN* had the opposite effect (β = −0.167, 95% CrI = [−0.343, 0.003], p(ROPE) = 0.208). Figure 2 illustrates the posterior of the average predictions of the model for words with different values of *Similarity* and *CLPN*.

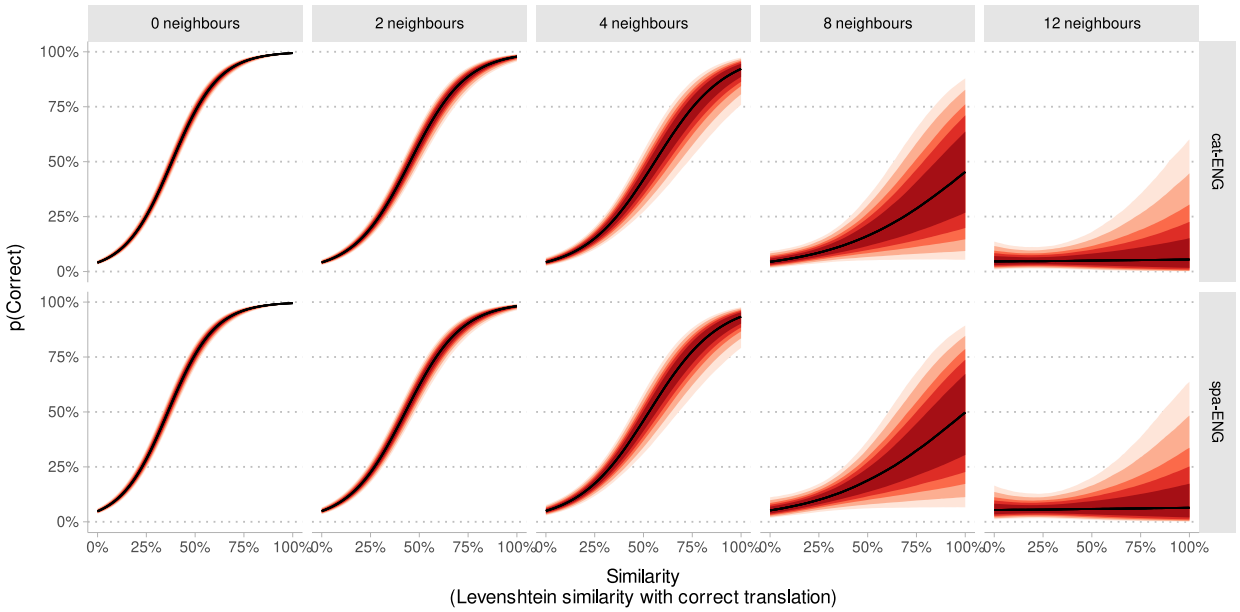
Table 3

Summary of participants' accuracy in the translation elicitation task across Experiments 1, 2, and 3.

		Accuracy (%)					Valid trials				
	N	M	SD	SE	Min	Max	M	N trials	SD	Min	Max
Experiment 1											
spa-ENG	27	0.16	0.05	0.03	0.09	0.29	96.37	2,602	2.88	87	98
cat-ENG	32	0.18	0.05	0.03	0.11	0.33	81.38	2,604	3.17	71	83
Experiment 2											
cat-SPA	21	0.48	0.05	0.11	0.38	0.59	79.14	1,662	3.14	72	82
spa-ENG	31	0.21	0.08	0.04	0.06	0.44	97.45	3,021	1.80	88	98
Experiment 3											
cat-ENG	32	0.20	0.05	0.03	0.10	0.28	82.75	2,648	0.44	82	83

Figure 2

Posterior model-predicted mean accuracy in Experiment 1. Predictions were generated from 4,000 posterior samples, extracted for different values of CLPN *(0, 2, 4, 8, 12) and* Similarity *(1-100). Predictions are plotted separately for English participants translating Catalan words, and for English participants translating Spanish words. Lines indicate mean predictions, and intervals indicate 95%, 89%, 78%, 67%, and 50% credible intervals (CrI).*



In Experiment 1, we show that phonological similarity between translation equivalents can support successful word translation through processes of homophonic translation, even in the absence of conceptual knowledge about the presented word. We tested two groups of adult British English native speakers in a translation task that involved words in Catalan or Spanish, two languages that the participants reported having no prior familiarity with. Participants benefited strongly from phonological similarity when the correct translation of the presented words in Catalan or Spanish had few English phonological neighbours with higher lexical frequency. As the number of higher-frequency English phonological neighbours increased, both participants' accuracy and the impact of phonological similarity approached zero. This suggests that the presentation of word forms in an unfamiliar language activates candidate

words in the listeners' native language, leading to correct translations when there is high phonological similarity between both words and few high-frequency phonological neighbours, even when the listener has no prior experience with the language.

Experiment 2

Results in Experiment 1 suggest that English native speakers were able to exploit the phonological similarity between unfamiliar words in Catalan and Spanish to provide accurate translations to English. This was observed despite English, a Germanic language, being relatively distant from Catalan and Spanish, two Romance languages. In comparison, Catalan and Spanish are typologically close languages that share many more cognates. In Experiment 2, we investigated whether listeners of an unfamiliar but typologically closer language benefit more strongly from phonological similarity when performing the same task as in Experiment 1. To this aim, we presented Spanish participants, who reported little-to-no prior familiarity with Catalan, with Catalan words.

Methods

We collected data from 33 Spanish native adults living in Spain (*Mean* = 21.85 years, *SD* = 3, range = 18-33, 28 female). Data collection took place from June 8th, 2020 to June 28th, 2020. Participants in Spain were contacted via announcements at the University campus(es), and were compensated €5 or an Amazon voucher for the same value. Participants gave informed consent before providing any data and the study was conducted in accordance with ethical standards of the Declaration of Helsinki and the protocol was approved by the Drug Research Ethical Committee (CEIm) of the IMIM Parc de Salut Mar (2020/9080/I).

Stimuli were the same list of Catalan stimuli as in Experiment 1. Procedure and data analysis were identical as in Experiment 1, with the only exception that the statistical model did not include the *Group* predictor, given that only one group (`cat-SPA`) participated in this experiment.

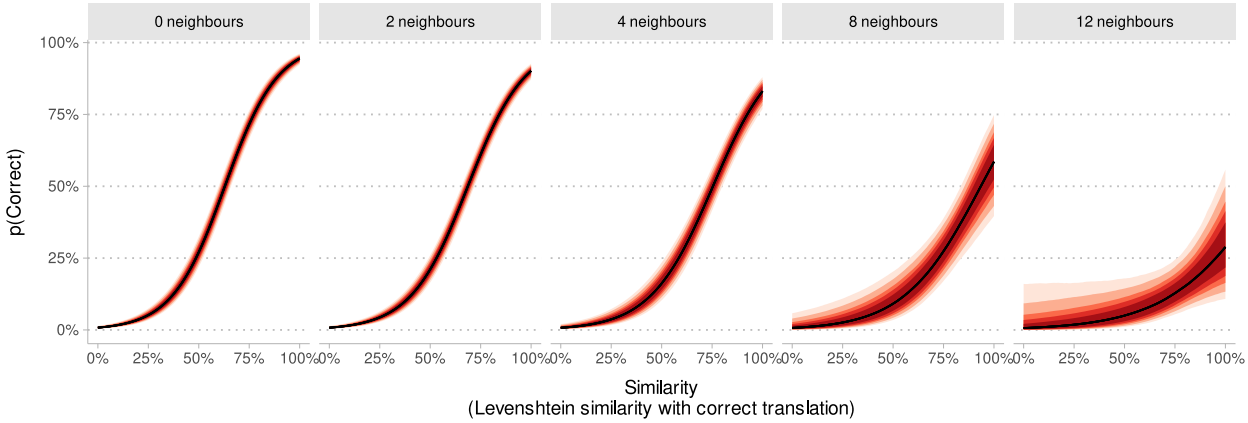
Results and discussion

We collected data for a total of 5,412 trials completed by 33 participants. We excluded trials in which participants did not enter any text (n = 44), in which a response in a language other than Spanish was provided (n = 51), in which participants did not provide a whole word (n = 7), and in which participants added comments to the experimenter (n = 1). In addition, we excluded data from participants that self-rated their oral and/or written skills in Catalan or any other second language as four or higher in a five-point scale (n = 22), were diagnosed with a developmental language disorder (n = 1), or did not contribute more than 80% of valid trials (n = 9). After applying trial-level and participant-level inclusion criteria, the resulting dataset included 1,662 trials provided by 42 participants. Responses given by participants were 5.6 characters long on average (*Median* = 5, *SD* = 1.6, range = 2-12).

Overall, participants responded less accurately to words with more CLPNs than to words with fewer CLPNs, regardless of the amount of phonological similarity between the presented word and its translation. This is indicated by the size of the regression coefficient of the two-way interaction between *Similarity* and *CLPN* (β = −0.159, 95% CrI = [−0.366, 0.032], p(ROPE) = 0.27). As anticipated, participants' performance benefited from an increase in *Similarity* (β = 1.905, 95% CrI = [1.721, 2.093], p(ROPE) = 0), while the number of *CLPN* had the opposite effect (β = −0.275, 95% CrI = [−0.618, 0.083], p(ROPE) = 0.145). Figure 3 illustrates the posterior of the average predictions of the model for words with different values of *Similarity* and *CLPN*.

Figure 3

Posterior model-predicted mean accuracy in Experiment 2. Predictions were generated from 4,000 posterior samples, extracted for different values of CLPN *(0, 2, 4, 8, 12) and* Similarity *(1-100). Lines indicate mean predictions, and intervals indicate 95%, 89%, 78%, 67%, and 50% credible intervals (CrI).*



In order to compare the results from Experiments 1 and 2 directly, we fit a model on the joint datasets from both experiments. This model included all predictors from the models presented in Experiments 1 and 2, and the predictor *Experiment* with levels `"Experiment 1"` and `"Experiment 2"`, and a three-way interaction between *Experiment*, *Similarity*, and *CLPN*. The *Experiment* variable was sum-coded as `-0.5 = Experiment 1` and `+0.5 = Experiment 2`. All two-way interactions between the three predictors were also included. The effect of *CLPN* on *Similarity* was equivalent in both groups, as indicated by the coefficient of the three-way way interaction (β = 0.111, 95% CrI = [−0.156, −0.047]). The posterior probability of this coefficient being equivalent to zero—*p*(ROPE)—was 0.417), that is, inconclusive. The coefficient of the interaction term between *Experiment* and *CLPN* was also equivalent to zero (β = −0.105, 95% CrI = [−0.487, 0.292], *p*(ROPE) = 0.343), suggesting that participants from Experiments 1 and 2 were affected by CLPN on a similar basis. Finally, the coefficient of the interaction between *Experiment* and *Similarity* was different from zero (β = −0.105, 95% CrI = [0.314,

0.786], p(ROPE) = 0), with a positive sign indicating that the increments in *Similarity* were associated with a larger increment in probability of correct translation in Experiment 2, compared to in Experiment 1: for an average value of *CLPN*, participants in Experiment 2 benefited more strongly from *Similarity* than participants in Experiment 1.

Experiment 2 was an extension of Experiment 1 to a population of monolinguals whose native language is typologically close to the presented language. We presented Catalan words to adult Spanish native speakers who were reportedly unfamiliar with Catalan. Our results indicate a similar pattern of results as those in Experiment 1: participants were able to provide correct translations of presented Catalan words, provided that the Catalan words shared some degree of phonological similarity with their Spanish translation, and that the number of phonological neighbours with higher lexical frequency was relatively low. In contrast with the results in Experiment 1, the positive impact of phonological similarity on participants' performance in Experiment 2 was larger. This suggests that participants in Experiment 2, who were native speakers of a typologically similar language (Spanish) to the presented language (Catalan), benefited more strongly from phonological similarity than participants in Experiment 1, who were native speakers of typologically less similar language (English) to the presented language (Catalan, Spanish).

However, an unexpected finding was that some trials with Catalan and Spanish words that had low similarity to their English translations were responded to with surprising accuracy by the English-speaking participants in Experiment 1. Table 4 shows a list of Catalan and Spanish words to which participants provided responses with 10% average accuracy despite low phonological similarity as indexed by the Levenshtein similarity score.

Table 4

List of items with unexpectedly high accuracy: the Levenshtein similarity score between the presented word (in Catalan or Spanish) and their correct English translation is zero, but participants, who are reportedly unfamiliar with the presented language, were on average >10% likely to guess the correct translation.

Translation	IPA	Accuracy (%)	SE
Experiment 1 (cat-ENG)			
cavall - horse	/kəˈbaʎ/ - /hɔːs/	17.14	6.37
llibre - book	/ˈʎi.βɾə/ - /bʊk/	17.14	6.37
camisa - shirt	/kaˈmi.za/ - /ʃɜːt/	16.67	6.21
poma - apple	/ˈpo.ma/ - /ˈæpl/	16.67	6.21
cama - leg	/ˈka.mə/ - /lɛg/	11.11	5.24
Experiment 2 (spa-ENG)			
pantalon - trousers	/paŋ.taˈlon/ - /ˈtraʊzəz/	77.42	7.51
naranja - orange	/naˈɾaŋ.xa/ - /ˈɒrɪnʤ/	41.94	8.86
leche - milk	/ˈle.t͡ʃe/ - /mɪlk/	35.48	8.59
toro - bull	/ˈto.ɾo/ - /bʊl/	33.33	8.61
libro - book	/ˈli.βɾo/ - /bʊk/	30.00	8.37
cebra - zebra	/ˈθe.bɾa/ - /ˈziːbrə/	29.03	8.15
pan - bread	/pan/ - /brɛd/	29.03	8.15
pollo - chicken	/ˈpo.ʎo/ - /ˈʧɪkɪn/	26.67	8.07
jirafa - giraffe	/xiˈɾa.fa/ - /ʤɪˈrɑːf/	20.69	7.52
perro - dog	/pe.ro/ - /dɒg/	16.13	6.61
pluma - feather	/plu.ma/ - /ˈfɛðə/	16.13	6.61
puerta - door	/pwer.ta/ - /dɔː/	16.13	6.61
pie - foot	/pje/ - /fʊt/	12.90	6.02
caballo - horse	/kaˈβa.ʎo/ - /hɔːs/	10.34	5.66
bocadillo - sandwich	/bo.kaˈdi.ʎo/ - /ˈsænwɪʤ/	10.00	5.48
globo - balloon	/ˈglo.βo/ - /bəˈluːn/	10.00	5.48
Experiment 3 (cat-SPA)			
fulla - hoja	/ˈfu.ʎə/ - /ˈo.xa/	30.43	9.59
ull - ojo	/uʎ/ - /ˈo.xo/	21.74	8.60
got - vaso	/ˈgɔt/ - /ˈba.so/	20.00	8.00
entrepa - bocadillo	/ˌen.trəˈpa/ - /bo.kaˈdi.ʎo/	13.04	7.02
mirall - espejo	/miˈɾaʎ/ - /esˈpe.xo/	12.50	6.75

It is possible that participants had prior knowledge of these words despite having reported little to no familiarity with the presented language as a whole. Spanish words percolate English speech with relative frequency, via different sources such as popular culture, songs, TV programs, etc. In addition, words from languages other than Spanish or Catalan, but with high similarity to the Spanish or Catalan words (e.g., cognates from Italian or French) might appear in English speech as well. Given that prior knowledge of specific words within the stimuli may have affected participants' answers, it is important to disentangle the extent to which participants' answers were driven by phonological similarity or prior knowledge they had about the meaning of the presented words. In Experiment 3, we sought to rule out this potential confound.

Experiment 3

Experiment 3 is a replication of Experiment 1 in which we collected additional data about participants' prior familiarity with the presented Catalan and Spanish words. We used this information to remove any responses in which participants reported prior knowledge with the meaning of the presented word from the final analysis.

Methods

We collected data from 64 British English native adults living in United Kingdom (*Mean* = 22.02 years, *SD* = 2.49, range = 18-26, 36 female). Data collection took place from October 22th, 2022 to October 23th, 2022. Participants were recruited via Prolific (5£ compensation) and SONA (compensation in academic credits), and gave informed consent before providing any data and the study was conducted in accordance with ethical standards of the Declaration of Helsinki and the protocol was approved by the University of Oxford Medical Sciences Inter-Divisional Research Ethics Committee (IDREC) (R60939/RE007). Participants were asked to complete the experiment using a laptop in a quiet place with good internet connection. Stimuli were the same lists of Catalan and Spanish stimuli as in Experiment 1.

The experiment was implemented online using Qualtrics (Qualtrics, Provo, UT). This platform was chosen to allow easier presentation of survey questions aimed to probe prior understanding of the presented words and participants' confidence ratings of their answers. With the exception of these additional questions, we attempted to replicate the procedure of Experiment 1 as closely as possible. The Spanish and Catalan audio stimuli used were identical the materials in Experiment 1. Participants were randomly assigned to the Catalan or Spanish lists. The Catalan list had 83 trials and the Spanish list had 99 trials. Participants first completed the consent form followed by the questionnaire about demographic status, language background and set up. They then proceeded to the experimental task.

In each trial, participants listened to the audio stimulus by clicking on the PLAY button. For comparability to the PsychoPy version, participants were only allowed to play the audio one time. Participants were explicitly told that they would be only allowed to listen once. The PLAY button vanished after one playthrough. Participants then had to answer three questions based on the audio they had heard on that trial. These questions were presented on the same page, directly below the audio player. They were first asked whether they knew the presented word (multiple choice—*yes*/*no*). Participants first completed 5 practice trials with English words as the audio stimulus (ambulance, cucumber, elephant, pear, turtle). The words were recorded by a female native speaker of British English. These trials acted as attention checks, as participants should always answer "yes" to the first question on prior word knowledge and be able to accurately transcribe the word they heard. Following the practice phase, participants completed the test phase where they heard either Spanish words or Catalan words.

Results and discussion

We collected data for a total of 6,016 trials completed by 64 participants. We excluded trials in which participants did not enter any text (n = 0), provided a response in a language other than English (n = 9), did not provide a whole word (n = 0), or added comments to the experimenter (n = 0). In addition, we excluded data from participants that self-rated their oral

and/or written skills in Catalan and Spanish, or any other second language as four or higher in a five-point scale (n = 2), were diagnosed with a language disorder (n = 0), or did not contribute more than 80% of valid trials (n = 1).

After applying trial-level and participant-level inclusion criteria, the resulting dataset included 5,888 trials provided by 63 participants. Of those trials, 3,145 were provided by 31 participants who listened to Catalan words, and 2,743 trials were provided by 32 participants who listened to Spanish words.

From the 3,145 total responses provided by English participants who listened to Catalan words, participants reported having prior knowledge of the presented Catalan words in 446 (14.18%) of them. In those responses, participants reported an average of 5.05 confidence in the 0-8 scale (SD = 1.94). For responses where no prior knowledge of the presented word was reported, average confidence was 1.13 (SD = 1.38). From the 2,743 total responses provided by English participants who listened to Spanish words, participants reported having prior knowledge of the presented Spanish words in 197 (7.18%) of them. In those responses, participants reported an average of 4.79 confidence in the 0-8 scale (SD = 1.82). For responses where no prior knowledge of the presented word was reported, average confidence was 1.25 (SD = 1.66). Before data analysis, responses where participants reported prior knowledge about the meaning of the presented Catalan or Spanish word were excluded from the dataset.

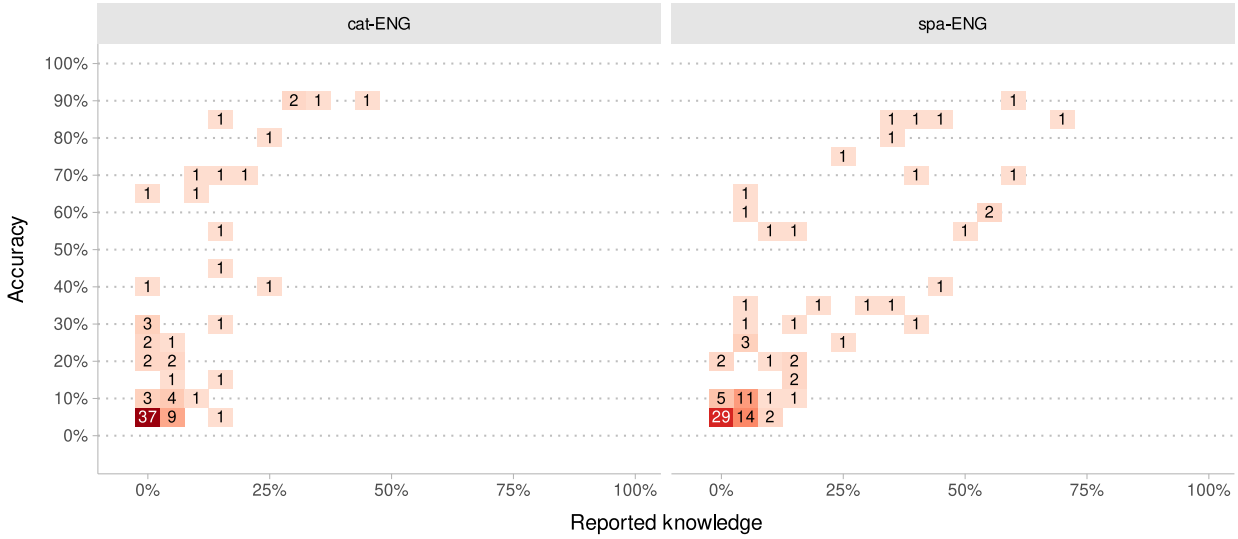
Responses given by English participants to Catalan presented words were 5.49 characters long on average ($Median$ = 5, SD = 1.71, range = 1-14), while their translations to Spanish responses were 5.39 characters long on average ($Median$ = 5, SD = 1.75, range = 1-20).

Overall, participants reported prior knowledge more often for the Spanish words that showed unexpectedly high accuracy in Experiment 1 than for words with expected accuracy (see Figure 4). Participants reported prior knowledge of Catalan words with unexpected accuracy in Experiment 1 as often as those with expected accuracy. This suggests that participants in Experiment 1 may have relied, to some extent, on their prior knowledge about

form-meaning mappings to correctly translate some Spanish words. To isolate such an effect of prior Spanish knowledge, we ran the same analysis as in Experiment 1 on the newly collected translations from Experiment 3, now excluding responses to words for which participants reported prior knowledge.

Figure 4

Catalan and Spanish prior word knowledge reported by English native participants in Experiment 3. The X-axis indicates the proportion of participants that reported prior knowledge with the word. The Y-axis indicates participants' accuracy (the proportion of participants that translated the word correctly). For visualisation purposes, data points have been aggregated so that numbers and color coding show the number of data points with identical accuracy and proportion of reported knowledge.

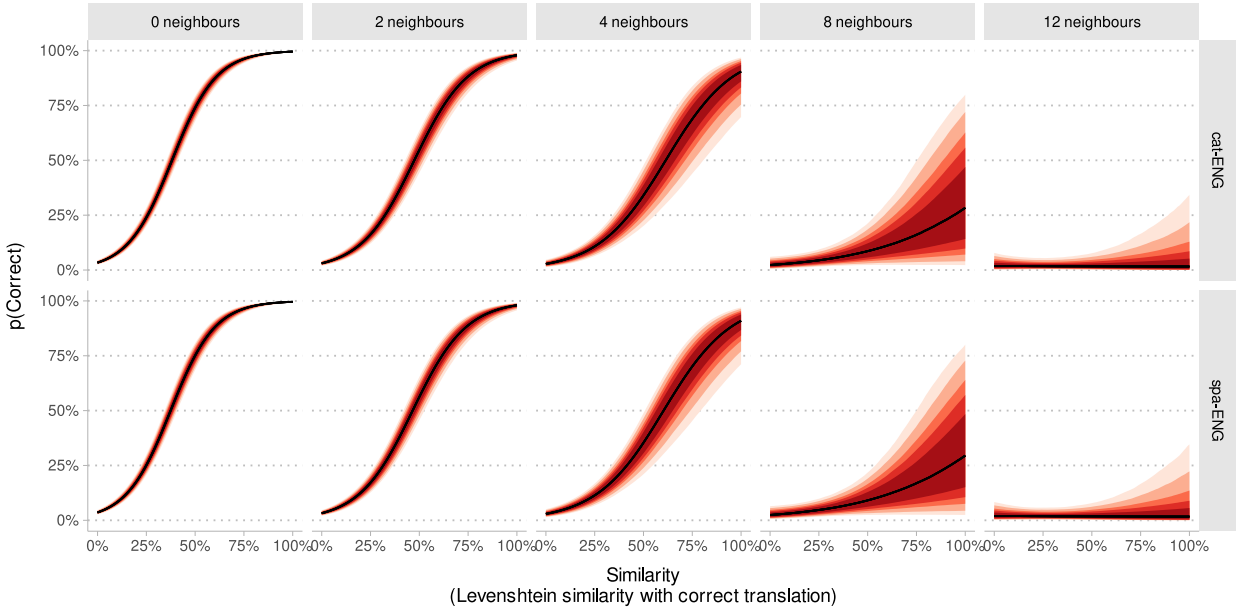


Participants translating Catalan words and participants translating Spanish words performed similarly, as indicated by the regression coefficient of *Group* (β = −0.063, 95% CrI = [−0.405, −0.122], p(ROPE) = 0.412). Overall, both groups of participants responded less accurately to words with more CLPNs than to words with fewer CLPNs, regardless of the amount of phonological similarity between the presented word and its translation. This is indicated by the size of the regression coefficient of the two-way interaction between

Similarity and *CLPN* (β = −0.248, 95% CrI = [−0.397, −0.122], p(ROPE) = 0.016). As anticipated, participants' performance benefited from an increase in *Similarity* (β = 1.532, 95% CrI = [1.406, 1.655], p(ROPE) = 0), while the number of *CLPN* had the opposite effect (β = −0.29, 95% CrI = [−0.532, −0.093], p(ROPE) = 0.02). Figure 5 illustrates the posterior of the average predictions of the model for words with different values of *Similarity* and *CLPN*.

Figure 5

Posterior model-predicted mean accuracy in Experiment 3. Predictions were generated from 4,000 posterior samples, extracted for different values of CLPN *(0, 2, 4, 8, 12) and* Similarity *(1-100). Predictions are plotted separately for English participants translating Catalan words, and for English participants translating Spanish words. Lines indicate mean predictions, and intervals indicate 95%, 89%, 78%, 67%, and 50% credible intervals (CrI).*



This follow-up experiment was designed to address concerns that participants in Experiments 1 and 2 might have managed to provide correct translations for some words thanks to having prior experience with those words. Findings from Experiment 3 suggest that while participants may have been relying on some prior knowledge when completing the task in Experiment 1, the observed effect of phonological similarity and CLPN can be found even

after excluding trials where participants report prior knowledge. Overall, our results suggest that participants were able to rely on the phonological similarity between the presented Catalan and Spanish words and their correct translation to guide word recognition in an unfamiliar language.

General discussion

Through a series of three experiments, we investigated how phonological similarity and its interaction with phonological neighbourhood density impact the dynamics of lexical activation and selection during processing of unfamiliar foreign words. Our findings support an account of homophonic translation in which this phenomenon results from listeners accommodating non-native phonological forms to their native phoneme inventory, resulting in the activation of native lexical representations. In Experiment 1, we found that English natives activate English word-forms that share some phonological similarity with auditorily presented Catalan and Spanish words (both unfamiliar languages to participants), and later engage dynamics of lexical competition and selection similar to those engaged in native speech processing. In Experiment 2, we extended these findings to a population of participants whose native language was typologically closer to the unfamiliar language presented. Spanish native adults who were unfamiliar with Catalan listened to a list of Catalan words. Spanish natives in Experiment 2 exploited phonological similarity even more strongly than English natives in Experiment 1, suggesting that the typological distance between the presented and target language is associated with a more robust facilitation effect of phonological similarity. In Experiment 3, we replicated the results from Experiment 1 after removing responses in which participants reported some prior knowledge about the meaning of the presented word, showing that participants' surprisingly good performance in the translation elicitation task was not driven by unreported prior experience with Catalan and Spanish.

The finding that participants were more likely to provide the correct translation when there was higher phonological similarity between the presented word and the correct

translation, despite being unfamiliar with the presented word, supports the idea that these phonological processes of homophonic translation can help facilitate the early stages of learning a new language by through capitalise on phonological similarity between the second language and their native language. Listeners are able to match foreign words to their native phonology and phonotactic despite the phonological form of the foreign word not being an exact match for a lexical item in their native language.

The mechanisms underlying this process of homophonic translation can be linked to established models of spoken word recognition. Listeners activate multiple candidate lexical items in their native language when they hear a spoken word and the candidate that sounds most similar to the presented word will receive the highest activation. When there are multiple candidates with similar levels of activation, they can act as competitors in the process of homophonic translation, as evidenced by the effect of higher-frequency phonological neighbours observed in our study.

Limitations and Future Directions

The present study extends previous research by providing a psycholinguistic account of homophonic translation. To investigate this, we presented participants with words from a language with which they had no prior experience and excluded from the analyses any words they had encountered previously. We observed consistent patterns across two participant groups with different native languages and varying degrees of similarity to the presented language, supporting our main conclusions. These findings provide insight into the dynamics of lexical processing in a foreign language when learners lack an established phonological or lexical representation of that language.

Nevertheless, the present study has several limitations. First, the measure of phonological similarity used here cannot fully capture all of the information available in the speech signal. In particular, prosodic features such as lexical stress, which were not included in our measure, may provide additional cues that help participants identify the correct translation

of the presented word forms. Second, phonological transcriptions such as X-SAMPA are abstract representations that disregard non-phonological variation (e.g., phonetic detail) and treat different symbols as categorically distinct phonemes, despite the fact that some phonemes share more phonetic features than others. Consequently, participants were likely able to exploit additional acoustic cues beyond those captured by our phonological similarity measure when producing correct translations. Future research should consider using measures of phonological similarity that incorporate finer-grained phonetic and prosodic information from the acoustic signal, as well as examining the relative contribution of these cues to homophonic translation.

Conclusions

Findings from the present study provide insights into the processing mechanisms underlying homophonic translation. Listeners were able to successfully exploit the phonological similarity between the presented foreign words and words in their native language to provide correct translations of the presented words despite no prior knowledge of the language. This facilitation effect of phonological similarity was stronger for participants whose native language was closer to the presented language. Importantly, participants only benefited from phonological similarity when the presented word had few higher-frequency phonological neighbours in their native language. Overall, the findings of the present study suggest that processing words in a non-native, unfamiliar language recruits mechanisms of lexical activation, selection and interference similar to those engaged during native-language word processing. These results indicate that core mechanisms of lexical processing operate even in the absence of an established lexical or phonological system for the target language, providing insight into the processes that may underlie the early stages of learning a second language.

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